Extreme Rapid Intensification of Hurricanes Otis (2023) and Patricia (2015)
A Machine Learning Diagnosis

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Objectives

Applying machine learning (ML) models to study different responses and skill in predicting of tropical cyclones (TC) rapid intensification (RI) has received more attention recently. In this study, we use ML models to examine the roles of environmental conditions on RI prediction. Specifically, our main objective is to examine the dominant environmental features at the RI onset moment versus 24 hours ahead of a TC track in RI predictions. Answering this questions can help explain the factors leading to the failure in predicting the extreme RI such as Hurricane Otis (2023) in all current operational models, while the same models could provide a successful RI prediction for Hurricane Patricia (2015).

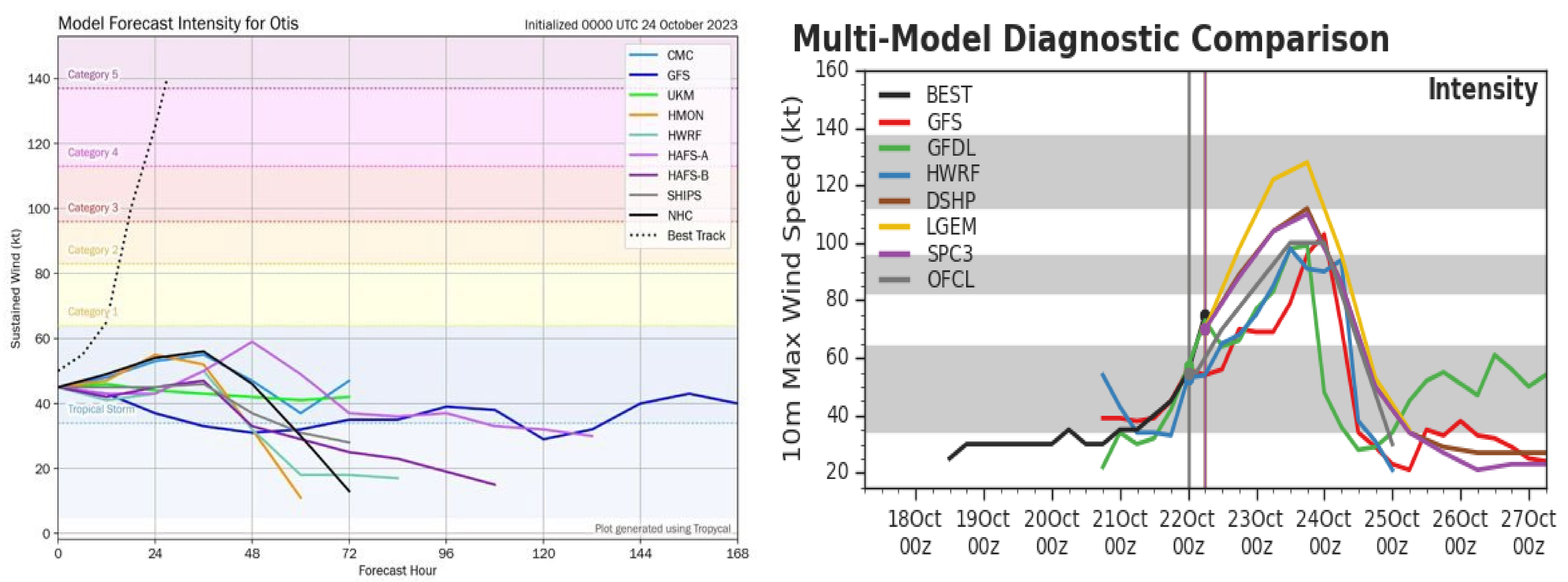

Figure 1. Real-time intensity forecasts for both Hurricane Otis (2023, left, Via. Tomer Burg) and Patricia (2015, right, Via CIRA). Dotted line represents observational wind speeds and colored lines represent several different NWP.

Machine Learning Models

To predict RI in TCs we employed 3 different classification models Random Forest, Gradient Boost, and Extreme Gradient Boosting that treated RI as a binary classification (yes/no) and built a probability of RI given several realizations ran. All data was trained using SHIPs diagnostic output from the HWRF, with training being 5856 models runs from 2011-2022 from all 3 major basins. The model was run in real time during the 2023 Atlantic and Eastern Pacific and the skill is evaluated for both 24-hour environments and initial environment (Fig.2).

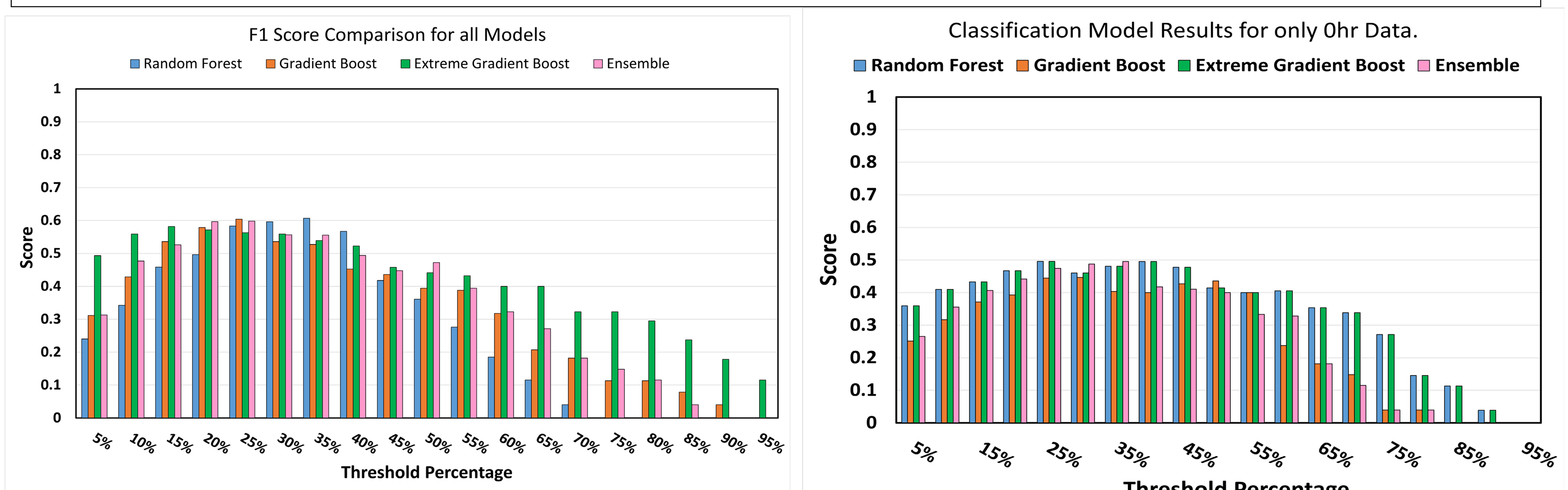

Figure 2. F1 score for a given forecasted RI probability using the 24-hour environments along TC track (Left) versus environments at the initial time only (Right).

Result # 1: Real-time Verification

- For 2023 season, the best classification thresholds for our ML models are in 30%-35%, reaching an F1 score of ~0.5 for RI prediction (Fig. 2);
- Using only initial environment gives worse performance, indicating that RI prediction requires environments ahead of TCs as well;
- ML model performance for RI prediction with only initial environments have less skill overall across threshold values; however their algorithm-to-algorithm skill is mostly the same, indicating the consistency in ML forecast skill.

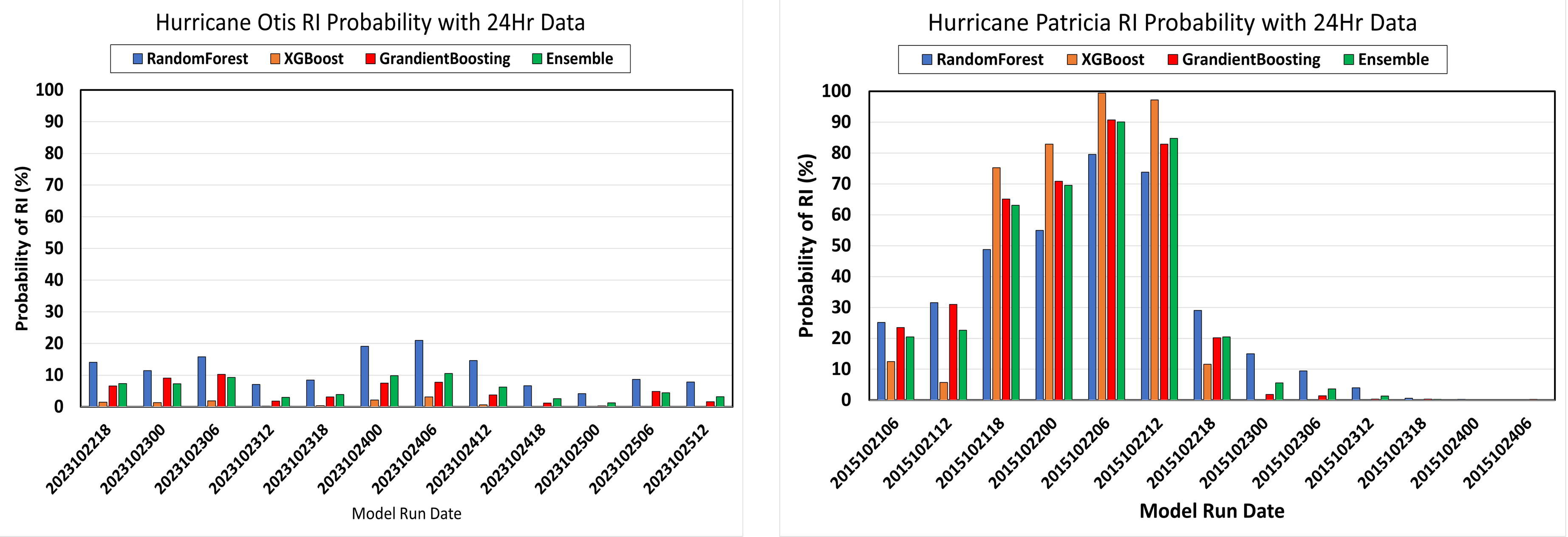

Figure 3. Classification model probability forecast of RI for Hurricane Otis (Left) and Hurricane Patricia (Right) at all cycles.

Result # 2: Otis's RI prediction

- ML model could not predict Otis's RI due to stronger shear, inconsistent and cooler SST, and slowly speed as compared to Hurricane Patricia (Fig. 3-4);
- NWP-based forecast can be very sensitive to shear and SST when trying to forecast RI, and our hypothesis is that the ML classification models may have inherited this trait from the NWP training set;
- To test this theory, we created a new set of storm data taking Hurricane Patricia's SST, shear, and storm translation speed and substituted them into Hurricane Otis's data and reran the classification models to see if our ML-based forecast of Oti's RI can be improved.

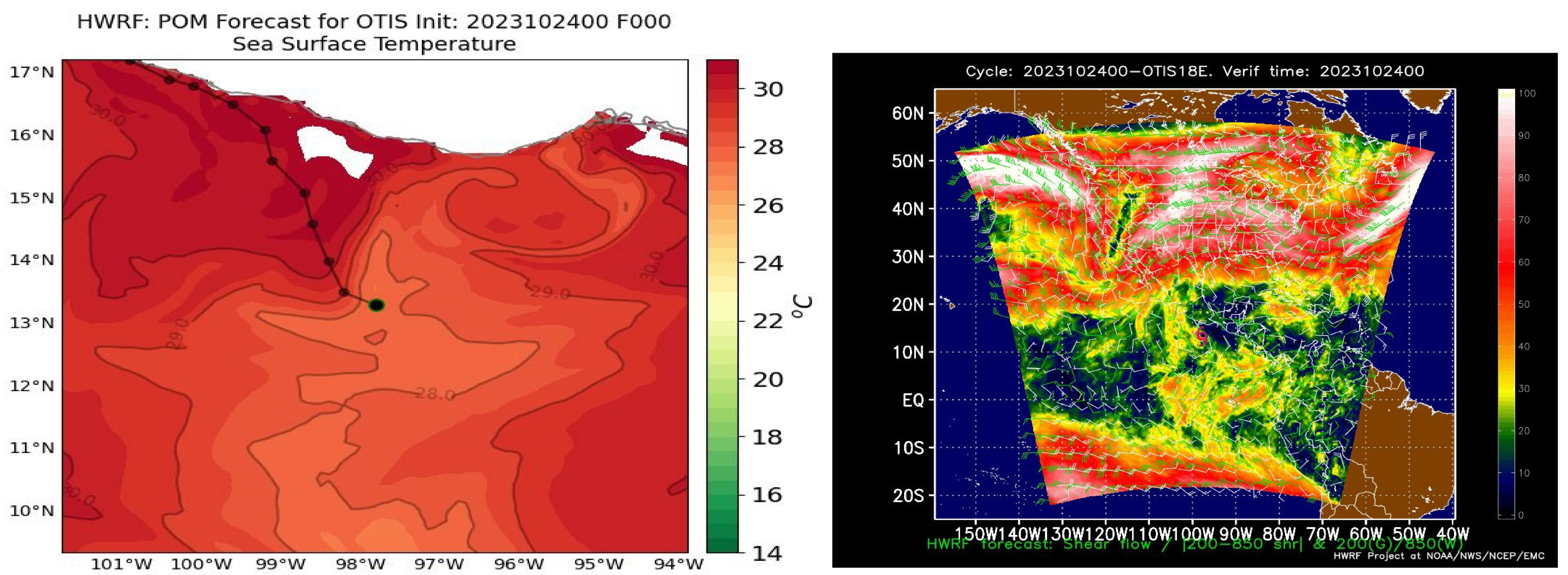

Figure 4. Sea Surface Temperatures from the operational HWRF forecast for (Left) and Deep Layer Shear (Right) from Hurricane Otis valid at 0000 UTC October 24, 2023.

Result # 3: Modified Otis Environment

- Using Hurricane Patricia's environment including SST, shear, and storm translation speed boosts the RI probability forecast for Otis to the 60-70%.
- All sensitivity experiments with our ML models show that the SST and shear played a large role in causing the suppressed RI forecast for both NWP and ML models;
- There is a possibility that our ML models may struggle with RI predictions for TCs in suboptimal environments such as Hurricane Otis, which is not included in the training dataset;
- Our additional sensitivity analyses revealed that ML models may have inherited strong SST and shear sensitivity that are present in the NWP.

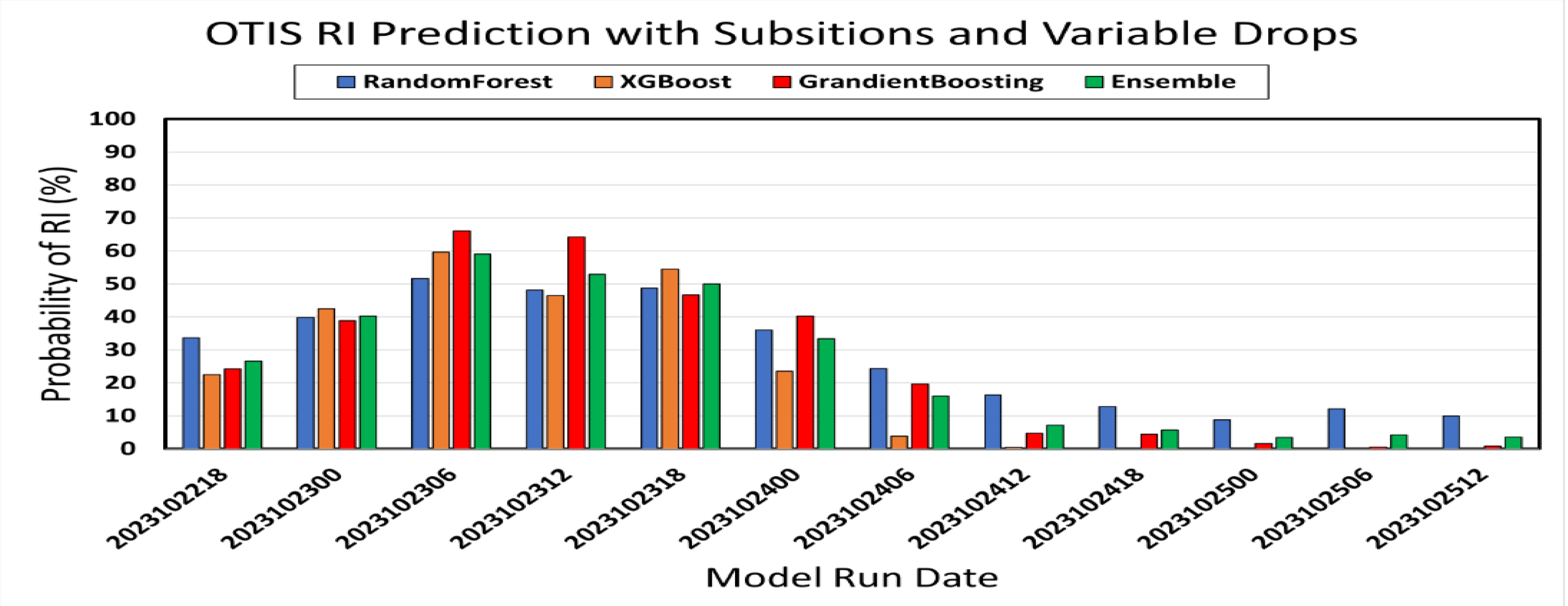

Figure 5. RI probability prediction from the ML classification models for Hurricane Otis, but using the Hurricane Patricia's environmental conditions.

Conclusions

ML classification models have shown the ability to produce some skillful RI prediction when proper storm environments are provided. Given storm environments, it is important to include model forecasted data out to 24 hours ahead of a TC track from physical-based model guidance, which can help increase the skill of RI prediction. Our real-time verification during 2023 season confirms the importance of environments ahead of a TC, thus showing that the future environment of the storm is important for RI predictions. However, when applying ML models in suboptimal environments such as Hurricane Otis, it is possible that ML models may have inherited a strong sensitivity to SST and shear from the training dataset that can reduce their performance in RI forecasts if these suboptimal environments are not included in the training dataset. This explains why our ML models failed to predict the RI of Hurricane Otis, when feeding the environmental conditions from HWRF's SHIP products.

References

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